

First name:

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SOLUTION

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Problem 1. Use a triple integral, thanks to the Divergence Theorem, to evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} \, dA$ for

$$\vec{F} = [x^2, 0, z^2],$$

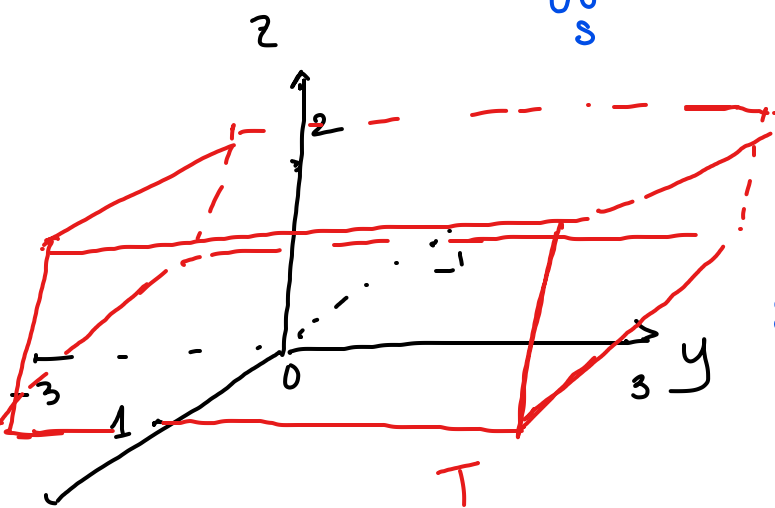
S is the surface of the box $|x| \leq 1$, $|y| \leq 3$, $0 \leq z \leq 2$.

[5 marks]

Solution.

By divergence theorem:

$$\oiint_S \vec{F} \cdot \vec{n} \, dA = \iiint_T \vec{\nabla} \cdot \vec{F} \, dV$$



For T :

$$\begin{aligned} -1 &\leq x \leq 1 \\ -3 &\leq y \leq 3 \\ 0 &\leq z \leq 2 \end{aligned}$$

So we get

$$\int_0^2 \int_{-3}^3 \int_{-1}^1 \vec{\nabla} \cdot \vec{F} \, dx \, dy \, dz$$

Now $\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} x^2 + \frac{\partial}{\partial y} 0 + \frac{\partial}{\partial z} z^2 = 2x + 2z$

Thus: $\int_0^2 \int_{-3}^3 \int_{-1}^1 2x + 2z \, dx \, dy \, dz = \int_0^2 \int_{-3}^3 \left[x^2 + 2zx \right]_{-1}^1 dy \, dz$

$$= \int_0^2 \int_{-3}^3 ((1)^2 + 2z(1) - (-1)^2 - 2z(-1)) dy \, dz = \int_0^2 \int_{-3}^3 4z \, dy \, dz$$

$$= \int_0^2 4zy \Big|_{-3}^3 dz = \int_0^2 (4z(3) - 4z(-3)) dz = \int_0^2 24z \, dz$$

$$= 12z^2 \Big|_0^2 = 12(2)^2 - 12(0)^2 = 48.$$

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Problem 2. Verify Stokes' Theorem for

$$\vec{F} = [y^3, -x^3, 0],$$

and S is the surface

$$x^2 + y^2 \leq 1, \quad z = 0.$$

Hint: To save time, you may directly use, if needed, the following facts:

1. S can be parameterized as $x = u \cos v$, $y = u \sin v$, for $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$, and in this set up, $\vec{N} = \vec{r}_u \times \vec{r}_v = u\hat{k}$.
2. The boundary ∂S , can be parameterized as $x(t) = \cos(t)$, $y(t) = \sin(t)$, $0 \leq t \leq 2\pi$.
3. The expression $\sin^4(t) + \cos^4(t)$ can be simplified to $\frac{3}{4} + \frac{1}{4}\cos(4t)$.

[5 marks]

Solution.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^3 & -x^3 & 0 \end{vmatrix} = \hat{k} \left(\frac{\partial}{\partial x}(-x^3) - \frac{\partial}{\partial y}(y^3) \right) = \hat{k}(-3x^2 - 3y^2) = -3(x^2 + y^2)\hat{k}$$

Using given parameterization:

$$\begin{aligned} \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} \, dA &= \int_0^{2\pi} \int_0^1 (\vec{\nabla} \times \vec{F}) \cdot \vec{N} \, du \, dv \\ &= \int_0^{2\pi} \int_0^1 -3(u^2 \cos^2 v + u^2 \sin^2 v) \hat{k} \cdot u \hat{k} \, du \, dv \\ &= -3 \int_0^{2\pi} \int_0^1 u^3 \, du \, dv = -3 \int_0^{2\pi} \left[\frac{u^4}{4} \right]_0^1 dv \\ &= -3(2\pi) \left(\frac{1}{4} \right) = -\frac{3\pi}{2} \end{aligned}$$

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Now:

$$\vec{r} = [\cos(t), \sin(t), 0] \Rightarrow \vec{r}' = [-\sin(t), \cos(t), 0]$$

$$\begin{aligned} \oint_{\partial S} \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} [\sin^3 t, -\cos^3 t, 0] \cdot [-\sin t, \cos t, 0] dt \\ &= \int_0^{2\pi} -\sin^4 t - \cos^4 t dt = - \int_0^{2\pi} \sin^4 t + \cos^4 t dt \\ &= - \int_0^{2\pi} \left[\frac{3}{4} + \frac{1}{4} \cos(4t) \right] dt = - \left[\frac{3}{4}t + \frac{1}{16} \sin(4t) \right]_0^{2\pi} \\ &= - \left[\frac{3}{4}(2\pi) + \frac{1}{16} \sin(4(2\pi)) - \frac{3}{4}(0) - \frac{1}{16} \sin(4(0)) \right] \\ &= -\frac{3\pi}{2} \end{aligned}$$

$$\text{As } \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} dA = \oint_{\partial S} \vec{F} \cdot d\vec{r}$$

Stokes' theorem has been verified!

MATH73235

Spring 2025

Quiz 5

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